

AN EXACT COMPLEMENT-NORM ATLAS FOR THE EXTERIOR ERDŐS-STRAUS CHANNEL

THE CLANKERS

ABSTRACT. We retain the original exterior residual R , its cofactor D , and the complementary divisor $v = a^2/u$. Every unit first-half exterior state satisfies $(p + w)^2 > 4(w + 1)^2(w - 1)$ for $w = \min(v, R, D)$. This yields a complete disjoint three-chart atlas whose outer parameter cutoff is $(p/2)^{2/3} + O(p^{1/3})$. The complement chart is an actual norm-divisor construction with the additional indispensable square-divisor availability condition. An original hard-prime state disproves transferring the preceding middle square-root cutoff to exterior cofactors. An explicit unit integer family establishes asymptotic sharpness in that stated domain. The universal existence assertion remains unproved.

1. SCOPE: COMPLEMENTING THE MIDDLE ATLAS WITHOUT ASSERTING EXISTENCE

The preceding middle-channel result proves a complete two-chart atlas with a square-root cutoff [4]. It does not prove that that atlas is occupied. The present continuation treats the other channel. It gives a three-chart exterior atlas indexed by an explicit cubic cutoff, preserves the original square-divisor availability in the norm chart, and disproves transferring the middle square-root bound unchanged to the exterior cofactors. The universal existence step remains unproved. In particular, a finite decision algorithm is not a witness-guaranteed algorithm.

The exterior normalization below is an explicitly marked presentation of the classical Type I parametrization; compare [1, §2, Proposition 2.2] and the recent divisor-parametrization comparison [2, §3]. The paired middle channel is Type II in the sense of [1, §2, Proposition 2.6]. Every algebraic identity used here is proved below in the retained marking. No historical-priority claim is made for the derived cutoff or atlas.

2. THE ORIGINAL EXTERIOR STATE AND ITS THREE ARITHMETIC PARAMETERS

Let $p, a, u \in \mathbb{Z}_{>0}$, with $p > 1$ and $p \equiv 1 \pmod{4}$, and retain the unit first-half source

$$p/4 < a < p/2, \quad \gcd(p, a) = 1, \quad R = 4a - p, \quad u \in \mathbb{Z}_{>0}, \quad u \mid a^2, \quad R \mid 4u + 1. \quad (1)$$

The cutoff theorem applies even when this integer parameter p is composite. When asserting a complete prime ES atlas, primality is imposed separately. Write $a = \prod_{\ell} \ell^{e_{\ell}}$ and retain every original exponent $\beta_{\ell} = v_{\ell}(u) - e_{\ell} \in [-e_{\ell}, e_{\ell}]$. Set

$$g = \gcd(a, u), \quad (h, r, s) = (g^2/u, u/g, a/g), \quad \kappa = (pr + s)/R. \quad (2)$$

Prime valuations prove that these are positive integers, $a = hrs$, $u = hr^2$, and $\gcd(r, s) = 1$. The gate in (1), after multiplication by the unit s modulo R , gives $R \mid pr + s$. The ordered return is

$$(x, y, z) = (a, hsk, phr\kappa), \quad \frac{4}{p} = \frac{1}{x} + \frac{1}{y} + \frac{1}{z}. \quad (3)$$

It is checked by multiplication. Its ordered inverse is

$$u = \frac{a^2}{Ry - pa}.$$

No denominator permutation is suppressed.

Retain the complement divisor and the exterior cofactor

$$\boxed{v = \frac{a^2}{u} = hs^2, \quad D = \frac{4u+1}{R} = \frac{r+\kappa}{s}.} \quad (4)$$

The second equality follows from $RsD = s(4hr^2 + 1) = (p+R)r + s = R(r+\kappa)$. Consequently

$$\boxed{pD + 1 = 4hr\kappa, \quad (p+R)^2 + 4v = 4vRD.} \quad (5)$$

Here $R, D \equiv 3 \pmod{4}$, and $\kappa > r$ because $R < p$ and $R\kappa = pr + s$. Thus

$$\boxed{p+R < 2vD.} \quad (6)$$

The case $h = 1$ would make -1 a square modulo $R \equiv 3 \pmod{4}$ through $R \mid 4r^2 + 1$. Indeed, some prime $q \equiv 3 \pmod{4}$ occurs to odd valuation in R ; the divisibility also gives $q \nmid 2r$ and $(2r)^2 \equiv -1 \pmod{q}$, contradicting Euler's criterion. Hence $h > 1$. Further, $\gcd(h, R) = \gcd(h, D) = 1$ by (1) and (5), so

$$\boxed{v \neq R, \quad v \neq D.} \quad (7)$$

This does *not* assert $\gcd(v, D) = 1$: the square factor s^2 may overlap D . For example, at $p = 37$, $(a, u, h, r, s, \kappa) = (12, 8, 2, 2, 3, 7)$ has $(v, D) = (18, 3)$.

3. A STRICT CUBIC CUTOFF

Theorem 3.1 (Exterior three-parameter bound). *For every state (1), put $w = \min(v, R, D)$. Then*

$$\boxed{(p+w)^2 > 4(w+1)^2(w-1).} \quad (8)$$

Let $C(p)$ be the largest positive integer w satisfying (8). Then

$$1 \leq C(p) < p, \quad \min(v, R, D) \leq C(p), \quad C(p) = (p/2)^{2/3} + O(p^{1/3}). \quad (9)$$

The constant and strict sign in (8) are part of the statement. The asymptotic notation is not used in the verifier: $C(p)$ is computed by exact integer comparison.

Proof. Put $t = \min(R, D)$. If $R = t$, (5) gives

$$(p+t)^2 \geq 4v(t^2 - 1).$$

If $D = t \leq R$, direct subtraction using (5) gives

$$(p+t)^2 - 4v(t^2 - 1) = (R-t)(4vt - 2p - R - t) \geq 0. \quad (10)$$

Indeed (6) implies $4vt - 2p - 2R > 0$; the last factor in (10) is that positive quantity plus $R - t$. Equality is possible only when $R = D = t$. Hence

$$p \geq 2\sqrt{v(t^2 - 1)} - t. \quad (11)$$

By (7), either $v \geq t+1$ or $v \leq t-1$.

In the first case $w = t$, and (11) gives $p \geq 2(w+1)\sqrt{w-1} - w$. Equality would require $R = D = w$, $v = w+1$, and $p+w = 2(w+1)\sqrt{w-1}$. Thus $w-1$ would be an integer square, whereas $w = R \equiv 3 \pmod{4}$ makes $w-1 \equiv 2 \pmod{4}$. Equality is impossible.

In the second case $w = v$ and $t \geq w+1$. The function $2\sqrt{w(t^2 - 1)} - t$ is strictly increasing for $t > 1$, so

$$p \geq 2w\sqrt{w+2} - w - 1.$$

This is strictly larger than $2(w+1)\sqrt{w-1} - w$. To verify the comparison, put $U = w\sqrt{w+2}$ and $V = (w+1)\sqrt{w-1}$. For $w \geq 2$,

$$U^2 - V^2 = w^2 + w + 1, \quad U + V \leq 2w(w+1),$$

so $U - V > 1/2$. This proves the strict inequality in both cases; squaring its positive sides proves (8).

The function

$$F(w) = 2(w+1)\sqrt{w-1} - w$$

is strictly increasing for $w > 1$. At $w = 1$, (8) holds, and at $w = p$ it fails since $(p+1)^2(p-1) - p^2 = p^3 - p - 1 > 0$. There is therefore one final true integer in $[1, p-1]$, obtained by binary search. The defining inequality is precisely $F(w) < p$. Moreover

$$F(w) = 2w^{3/2} - w + O(\sqrt{w}) \quad (w \rightarrow \infty).$$

Maximality gives $F(C(p)) < p \leq F(C(p) + 1)$, and the displayed expansion at $C(p)$ and $C(p) + 1$ gives

$$p = 2C(p)^{3/2} + O(C(p)).$$

In particular $C(p) \asymp p^{2/3}$; dividing by $2C(p)^{3/2}$ and taking the $2/3$ power then yields

$$(p/2)^{2/3} = C(p) + O(C(p)^{1/2}) = C(p) + O(p^{1/3}),$$

which is (9) with the stated leading constant. $\square$

The residue hypothesis in the strict form is essential. If it is removed, then

$$(p, a, u, R, D, v, w) = (19, 6, 6, 5, 5, 6, 5)$$

satisfies the positive first-half divisor data but gives equality rather than strict inequality:

$$(p + w)^2 = 24^2 = 576 = 4 \cdot 6^2 \cdot 4 = 4(w + 1)^2(w - 1).$$

This is a boundary counterexample to the hypothesis-free statement, not an Erdős–Straus counterexample.

For a prime p , the least-nonresidue lower bound from the preceding work gives $v = hs^2 \geq h \geq \nu_p$. This may prune the complement chart, but it does not replace the requirement that that chart actually contain a divisor satisfying its congruence and availability condition.

4. THREE DISJOINT CHARTS WITH COMPLETE INVERSES

Define the root-divisibility kernel

$$K(v) = \prod_{\ell^e \parallel v} \ell^{\lceil e/2 \rceil}.$$

Then $v \mid a^2$ is equivalent to $K(v) \mid a$, prime by prime. For the following atlas let $p \equiv 1 \pmod{4}$ be prime and $C = C(p)$.

Theorem 4.1 (Complete original exterior atlas). *Every original first-half exterior state belongs to exactly one of the following charts, and every chart entry reconstructs such a state.*

Direct residual chart. Retain

$$3 \leq R \leq C, \quad R \equiv 3 \pmod{4}, \quad a = (p + R)/4, \quad u \in \mathbb{Z}_{>0}, \quad u \mid a^2, \quad R \mid 4u + 1.$$

Compute $D = (4u + 1)/R$, $v = a^2/u$, and retain $R \leq D$, $R < v$. The map and inverse are (1)–(4).

Reciprocal chart. Retain

$$3 \leq D \leq C, \quad D \equiv 3 \pmod{4}, \quad B_D = (pD + 1)/4, \\ w \in \mathbb{Z}_{>0}, \quad w \mid B_D^2, \quad w < B_D, \quad D \mid w + B_D.$$

Set

$$g = \gcd(B_D, w), \quad h = g^2/w, \quad r = w/g, \quad \kappa = B_D/g, \quad s = (r + \kappa)/D, \quad a = hrs, \quad u = w.$$

Compute $R = 4a - p$, $v = hs^2$, and retain $D < R$, $D < v$.

Complement–norm chart. Retain

$$2 \leq v \leq C, \quad R \mid p^2 + 4v, \quad R \equiv 3 \pmod{4}, \quad v < R < p, \quad R \equiv -p \pmod{4K(v)}. \quad (12)$$

Put

$$a = (p + R)/4, \quad u = a^2/v, \quad D = (4u + 1)/R, \quad (13)$$

and retain $v < D$. Formula (2) then gives the complete original marking and (3) its ordered denominators. The inverse reads the original complement $v = a^2/u$ and residual R .

Proof. For the direct chart, $p + R \equiv 0 \pmod{4}$ makes $a = (p + R)/4$ integral; $0 < R \leq C < p$ gives $p/4 < a < p/2$; and primality of p gives $\gcd(p, a) = 1$. The two displayed divisor conditions are exactly the remaining source conditions, so

$$(R, u) \mapsto ((p + R)/4, u), \quad (a, u) \mapsto (4a - p, u)$$

are mutually inverse on the retained direct region. In particular the full u -fibre is retained; the outer residual R alone is not treated as a state.

For the reciprocal chart put $B = B_D$ and $g = \gcd(B, w)$. Prime valuations in $w \mid B^2$ prove that

$$h = \frac{g^2}{w} \in \mathbb{Z}_{>0}, \quad w = hr^2, \quad B = hr\kappa, \quad \gcd(r, \kappa) = 1.$$

The inequality $w < B$ is exactly $r < \kappa$. Since $pD + 1 = 4B$, the odd integer D is coprime to B , hence to $g = hr$. The condition $D \mid w + B = g(r + \kappa)$ therefore gives $D \mid r + \kappa$, so $s = (r + \kappa)/D$ is a positive integer. If a prime divided both r and s , the identity $Ds = r + \kappa$ would make it divide κ , contrary to $\gcd(r, \kappa) = 1$; thus $\gcd(r, s) = 1$.

Now

$$R\kappa = (4hrs - p)\kappa = s(4hr\kappa) - p\kappa = s(pD + 1) - p\kappa = pr + s > 0,$$

and

$$DRs = R(r + \kappa) = Rr + R\kappa = (R + p)r + s = s(4hr^2 + 1).$$

Cancelling the positive integer s proves $DR = 4u + 1$ for $u = w = hr^2$. Furthermore

$$D(p - 2a) = pD - 2hr(r + \kappa) = 4hr\kappa - 1 - 2hr(r + \kappa) = 2hr(\kappa - r) - 1 > 0.$$

Thus $R > 0$, $p/4 < a < p/2$, and, because p is prime and $0 < a < p$, $\gcd(p, a) = 1$. We have recovered every condition in (1), not merely the ordered reciprocal identity. Conversely, for an original state in this chart, $B_D = hr\kappa$ and $w = u = hr^2$, while

$$d \mid \gcd(r, \kappa) \implies d \mid (R\kappa - pr) = s \implies d = 1,$$

because $\gcd(r, s) = 1$. Therefore

$$\gcd(B_D, w) = hr \gcd(\kappa, r) = hr = g.$$

The displayed formulas therefore recover $h, r, \kappa, s, a, u, R, D, v$ exactly. The complementary reciprocal divisor would give $\kappa < r$ and is not silently identified with this orientation. All divisors w in this chart are positive; admitting signed divisors would destroy the reconstructed positive state.

For (12), let $d \mid \gcd(R, 4v)$. The norm divisibility $R \mid p^2 + 4v$ gives $d \mid p^2$. Since p is prime and $0 < R < p$, the prime p cannot divide d , and hence $d = 1$. Thus $\gcd(R, 4v) = 1$. The residue modulo $4K(v)$ gives $K(v) \mid a$, equivalently the actual integer availability $v \mid a^2$, so (13) defines $u \in \mathbb{Z}_{>0}$. Finally

$$4v(4u + 1) = (p + R)^2 + 4v \equiv 0 \pmod{R}.$$

Cancelling the proved unit $4v$ gives the exterior gate and makes $D = (4u + 1)/R$ a positive integer. The range gives $p/4 < a < p/2$; since p is prime it also gives $\gcd(p, a) = 1$. Thus every condition in (1) has been reconstructed, and gcd normalization applies. Conversely every original state has $R \mid p^2 + 4v$ by (5), and its $v \mid a^2$ gives exactly the residue modulo $4K(v)$.

The smallest of R, D, v is at most C by Theorem 3.1. The grade v cannot tie either of the other parameters. If $R \leq D$, then either $R < v$, which is the direct chart, or $v < R \leq D$, which is the norm chart. If $D < R$, then either $D < v$, which is the reciprocal chart, or $v < D < R$, which is the norm chart. Thus the three rules are disjoint and exhaustive. More precisely, an $R = D$ tie for the minimum, $R = D < v$, is assigned to the direct chart, while $v < R = D$ belongs to the complement–norm chart. Both cases occur: $(p, a, u, R, D, v) = (13, 4, 2, 3, 3, 8)$ is direct, whereas $(5, 2, 2, 3, 3, 2)$ is norm. $\square$

Primality is not cosmetic in the generative atlas. If it is dropped without adding replacement coprimality conditions, the direct data

$$(p, R, a, u, D, v) = (21, 3, 6, 2, 3, 18)$$

pass the displayed arithmetic tests but have $\gcd(p, a) = 3$; the reciprocal data $(p, D, B_D, w) = (93, 3, 70, 20)$ reconstruct $a = 30$ with the same defect; and the norm data $(p, v, R) = (25, 5, 15)$ fail the cancellation needed for $R \mid 4u + 1$. The cutoff and the classification of already-valid source states do not use primality, but these three parameter-side converse maps do.

A correction to an overly strong coprimality shortcut. One may cancel R against v in (12), because it follows there from the prime parameter and the norm divisibility. One may not assert $\gcd(v, D) = 1$ in the original state. The $p = 37$ example above has a common factor three. The reciprocal construction retains that factor through s .

Candidate size and the surviving existence question. There are at most $(C + 1)/4$ values in each of the first two lists and $C - 1$ values of v in the third. Exact finite candidate counts are obtained by summing $\tau(a^2)$, $(\tau(B_D^2) - 1)/2$, and $\tau(p^2 + 4v)$ over their respective lists, then applying the displayed original conditions. These are factorization and divisor-enumeration bounds, not polynomial-time claims. The preceding middle two-chart atlas has square-root-sized outer lists. Together the two results decide the complete prime ES source at each given p . They still do not prove that its result is nonempty for every p .

5. AN ACTUAL HARD PRIME AND AN ASYMPTOTICALLY EXTREMAL INTEGER FAMILY

At

$$\boxed{p = 48049 \equiv 169 \pmod{840}},$$

complete trial division through 219 proves primality. The original exterior state is

$$\boxed{(a, u, R, D, v, h, r, s, \kappa) = (12090, 24180, 311, 311, 6045, 6045, 2, 1, 309)}. \quad (14)$$

It gives

$$\boxed{\frac{4}{48049} = \frac{1}{12090} + \frac{1}{1867905} + \frac{1}{179501934690}}.$$

Here $4u + 1 = 311^2$, so the exterior cofactors genuinely share all their factors. The preceding hard-prime middle cutoff is 247. Both R and D exceed it. Thus the middle square-root bound cannot be transferred unchanged to these exterior parameters; the surviving map is Theorem 4.1, not a false identification of the two channels.

For every integer $n \geq 2$, set

$$\begin{aligned} h &= 4n^2 - 2, & r &= n, & s &= 1, & R &= D = 4n^2 - 1, & \kappa &= 4n^2 - n - 1, \\ p &= 16n^3 - 4n^2 - 8n + 1, & a &= hn, & u &= hn^2. \end{aligned} \quad (15)$$

Then $R^2 = 4hn^2 + 1$, $p = 4a - R$, and $R\kappa = pn + 1$. Also $\gcd(p, a) = 1$ since $\gcd(R, h) = \gcd(R, n) = 1$. The strict first-half inequalities and the ordered identity follow directly. Thus these are actual unit integer sources, not arbitrary real triples satisfying an inequality. Here

$$\min(v, R, D) = h = 4n^2 - 2, \quad \frac{h}{(p/2)^{2/3}} \longrightarrow 1.$$

In fact $C(p) = h$ at every member of (15). Writing $w = h$, its parameter is $p = 2w\sqrt{w+2} - w - 1$, while

$$2(w+2)\sqrt{w} - w - 1 > 2w\sqrt{w+2} - w - 1 = p.$$

The left side is the bounding function at $w + 1$. Thus (8) holds at h and fails at $h + 1$. The integer cutoff itself is attained, including the prime examples $p = 97, 929$. The exponent $2/3$ and leading coefficient in the general integer-domain cutoff cannot be decreased uniformly for these three parameters. At $n = 2, 4$, the parameters $p = 97, 929$ are prime. The complete congruence classification is

$$p(n) \equiv 289 \pmod{840} \iff n \equiv 24 \text{ or } 164 \pmod{210}. \quad (16)$$

Indeed reduction modulo 8, 3, 5, 7 gives respectively $n \equiv 0 \pmod{2}$, $n \equiv 0, 2 \pmod{3}$, $n \equiv 4 \pmod{5}$, and $n \equiv 3 \pmod{7}$; the Chinese remainder theorem gives exactly the two classes in (16). No infinitude of prime values of the cubic is asserted; consequently the asymptotic sharpness claim is for the stated unit integer domain, not for a proved infinite prime subfamily.

The least complement grade is not automatically occupied. At $p = 2521$, $v = 11$ requires $R \mid 6355485$ and $R \equiv 31 \pmod{44}$, $0 < R < p$. The factorization $6355485 = 3^2 \cdot 5 \cdot 141233$ gives the possible divisors below p as 1, 3, 5, 9, 15, 45, none in that class. For completeness, $141233 - 1 = 2^4 \cdot 7 \cdot 13 \cdot 97$ and the Lucas test with base 3 has

$$3^{141232} \equiv 1 \pmod{141233},$$

while the residues at exponents $141232/q$, for $q = 2, 7, 13, 97$, are 141232, 85064, 78498, 79135; each residue minus one is coprime to 141233. Thus 141233 is prime and the divisor list is complete. This antecedent extremal failure remains an explicit negative control [3].

6. STATUS, PROVENANCE AND REPLAY BOUNDARY

This continuation gives a complete exterior counterpart to the previous middle atlas. It does not close the unbounded existence step. The three original sources in Theorem 4.1 may still, logically, all be empty at a prescribed prime. A proof of a nonzero original coefficient, or a genuine prime-parameter cutoff with its entire exceptional domain certified, is still required before treating the proposed Turn 8 existence algorithm as established.

The main standard-library verifier compares every original first-half exterior state with the three-chart atlas at every $p \equiv 1 \pmod{4}$ prime through its recorded bound. The separate checker imports neither it nor predecessor code and enumerates primitive h, r, s directly. Both verify the hard example (14), the source-domain counterexamples, and the original integer family (15). The supplied execution receipt records actual outcomes, not outcomes inferred from the existence of source files. Replaying previous Turn 6–7 files repairs the downloadable handoff; it is not a new independent review of their proofs. No universal ES theorem, whole-history audit, Lean build, new overall ES verification range or historical-priority determination is claimed.

REFERENCES

- [1] Elsholtz, C., & Tao, T. (2013). Counting the number of solutions to the Erdős–Straus equation on unit fractions. *Journal of the Australian Mathematical Society*, 94(1), 50–105. [arXiv:1107.1010](#), §2, Propositions 2.2 and 2.6. Classical parametrizations; no existence theorem is imported.
- [2] Bello-Hernández, M., Benito, M., & Fernández, E. (2026). *A divisor parametrization for the Erdős–Straus conjecture*. [arXiv:2606.10922v1](#), §3. This is used as a recent literature comparison for the standard Type I/II forms, not as a source for the cubic cutoff proved here.
- [3] The Clankers. (2026). *Turn 6: pointwise localization of the complete source*. Preserved supplied file `es_turn6_full_shell_localization_20260919/core.tex`, sections 2, 4–8. Original marked returns, necessary bounds and the $p = 2521$ extremal obstruction.
- [4] The Clankers. (2026). *A sharp two-chart middle-selector cutoff and an original-grade obstruction for Erdős–Straus*. Preserved supplied file `es_turn7_middle_cutoff_20260919/core.tex`, sections 1–5. The middle atlas is a complete decision domain, not universal occupancy.